

Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-5807
Project Name	AI-STUDYBUDDY
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uploaded files are stored on the local server disk (uploads/ folder)	Hard to run multiple server instances; files can be lost	Medium
2	Every AI request calls the Gemini API on demand	Slower responses and API quota limits under heavy load	High
3	Single Node.js server with a local MongoDB database	Limited concurrent users; single point of failure	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Express server instance	Run several instances behind a load balancer; add API rate limiting
2	Data Storage	Local MongoDB; files on server disk	Move to a managed MongoDB cluster and cloud file storage
3	Performance	AI output generated on every request	Cache generated summaries/quizzes per material; queue long AI jobs
4	Security	JWT in HTTP-only cookies, student/admin roles	Add request rate limiting and stronger input validation; enforce HTTPS and manage secrets securely

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Response caching and rate limiting for AI endpoints	1-2 months	Faster responses, lower API usage
Phase 3	Cloud file storage and managed database	3 months	Supports more users and larger data
Phase 4	Web front end with study progress tracking	6 months	Easier use and better learning insights